

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMPUTER SCIENCE

COMPUTER SCIENCE 4021/1
PAPER 1 November 2023
1 hour 30 minutes

Candidate Name: _____

Centre Number: _____ **Candidate Number:** _____

TIME ALLOCATION & INSTRUCTIONS TO CANDIDATES

Theory of hardware, binary logic, data security, and pseudocode algorithms.

Answer all compulsory questions.

SECTION A

1. (a) Convert the 8-bit binary number 10011100 into its denary (decimal) equivalent. (b) Draw the truth table for a 2-input NAND gate. (c) State two differences between RAM and ROM. *[Binary Logic & Systems]* [6]
- (a) Convert 10011100 to denary. [2]
- (b) Provide NAND gate truth table. [2]
- (c) Contrast RAM and ROM. [2]